

Institutions, Culture, and Antecedents

Week 4

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Today

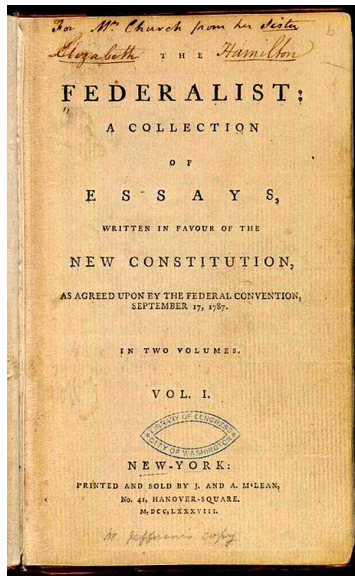
Last week asked whether institutions matter and how we can measure them. Today asks where they come from. Four answers, in order of increasing difficulty.

1. **They are designed.** Madison, writing mechanism design in 1787.
2. **They need an enforcer.** Johnson and Koyama on capacity and constraints.
3. **They get reallocated.** Coase on why the initial allocation matters and Lamoreaux on an America that kept taking property and growing anyway.
4. **They live in people's heads.** Bazzi, Fiszbein and Gebresilasse on the frontier.

The last paper will bring us back to a figure you saw last week.

Designing the rules

The document



First edition, 1788, from the Library of Congress. Eighty-five newspaper essays written to argue New York into ratifying, published under the name Publius. Number 10 is Madison's, and it is the one economists keep coming back to.

The problem Madison is solving

"AMONG the numerous advantages promised by a well-constructed Union, none deserves to be more accurately developed than its tendency to break and control the violence of faction."

"By a faction, I understand a number of citizens, whether amounting to a majority or a minority of the whole, who are united and actuated by some common impulse of passion, or of interest, adverse to the rights of other citizens, or to the permanent and aggregate interests of the community."

Note what the definition covers. A **majority** can be a faction. The threat to property is not only the king, and Week 3's expropriation risk has a democratic form.

Federalist No. 10, 1787.

Two cures, and why the first is worse than the disease

"There are two methods of curing the mischiefs of faction; the one, by removing its causes; the other, by controlling its effects."

"Liberty is to faction what air is to fire, an aliment without which it instantly expires. But it could not be less folly to abolish liberty, which is essential to political life, because it nourishes faction, than it would be to wish the annihilation of air, which is essential to animal life, because it imparts to fire its destructive agency."

"The inference to which we are brought is, that the CAUSES of faction cannot be removed, and that relief is only to be sought in the means of controlling its EFFECTS."

The second cure, making everyone think alike, he dismisses as impracticable so long as reason is fallible and people are free to use it.

The mechanism

- A minority faction is handled by voting. The hard case is a **majority** faction, which can sacrifice both the public good and the rights of others while keeping the forms of popular government.
- His answer is a **republic** rather than a pure democracy, and then a large one. Enlarge the sphere and no single interest spans a majority, so a majority can only be assembled by coalition.
- Then split the jurisdictions. *"The great and aggregate interests being referred to the national, the local and particular to the State legislatures."*

Read it as an economist. He states an incentive problem, rejects two candidate mechanisms on cost grounds, and derives a third from the structure of the problem. This is mechanism design, written in 1787.

What this frame does and does not settle

Institutions do not only evolve. Sometimes they are **built**, on purpose, by people who can say exactly which failure they are engineering against.

- That is a real answer to “where do institutions come from,” and it is the one most students reach for first.
- But a design on paper is not yet a working rule. Somebody has to be able to **enforce** it, across a territory, against people who would rather it were not enforced.
- Which costs money, and requires an organization capable of collecting it.

Building a state that can enforce them

The twin revolution

- As late as 1800 most of the world was poor and political authority almost everywhere was **dispersed and fragmented**.
- Preindustrial states rarely managed to tax more than about **5 percent** of output. Today OECD states take **20 to 40 percent**.
- So the modern economy and the modern state arrived together. That is a correlation.

The word “state” only acquired its modern meaning in English at the end of the sixteenth century. That marks the arrival of a thing that had not previously existed.

Johnson and Koyama (2017), “States and Economic Growth: Capacity and Constraints.” The survey I use to organize this literature, so treat me as an interested party.

Half of the twin revolution, from Week 1

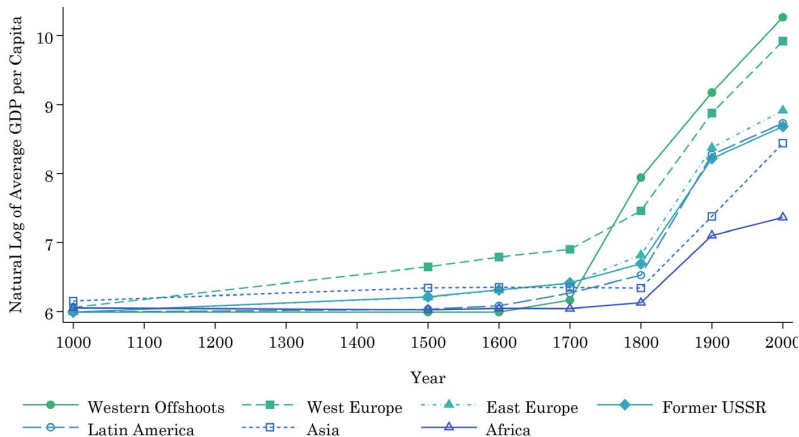
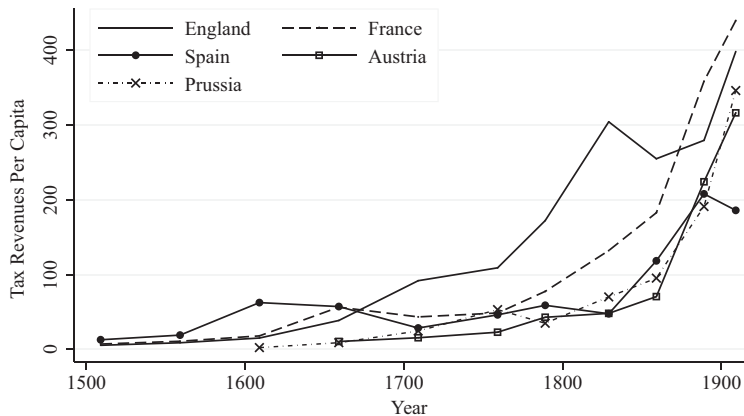


Fig. 1. Relative economic prosperity of different regions of the world from 1000 to 2000 CE.

Nunn (2020), *Science* 367(6485), Figure 1. The fact we opened the course with. Now look at what was happening to states over the same centuries.

The other half



Log tax revenues per capita for five European powers. Johnson and Koyama (2017), Figure 1, from Karaman and Pamuk (2013). Flat for two centuries, then the same hockey stick.

What the word meant before

- Latin *status* meant condition or standing. The *status regis* was the standing of the king, the *status regni* the condition of the realm. The word named a condition of something rather than a thing in its own right.
- Through the middle ages it referred to the ruler's own standing and magnificence. His estate, his position, his pomp. English keeps the fossils in "lying in state", "a state occasion", "stately" and "estate" itself.
- Machiavelli sits on the hinge. *Il Principe* opens on *tutti li stati*, and its recurring imperative is *mantenere lo stato*, to maintain the state. What the prince maintains is his own grip and dominion.

So when a medieval writer says a prince must maintain his state, he means his grandeur and his hold on power. Nobody could lend to that state, because there was no entity on the other side of the loan.

Skinner (2009), "A Genealogy of the Modern State."

The double abstraction

When a state exists

Authority becomes a **state** once it is separable from the ruler and from the ruled at the same time. The king can act against it and it outlives his death. And it is something other than the assembled citizens.

- **Bodin (1576)**. Sovereignty is power both absolute and **perpetual**. The perpetuity does the work, because it detaches the power from whoever currently holds it.
- **Hobbes (1651)**. The commonwealth is an artificial person, one person whose acts the multitude has authorized.
- **The English lawyers**. They arrive by another route, the King's Two Bodies. The Crown becomes an office with a legal personality of its own, separate from the body of the man wearing it.

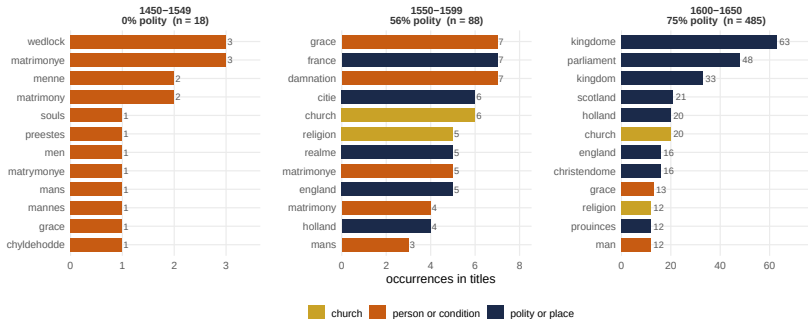
An office that outlives its holder can owe money. A man owes it only until he dies or decides otherwise. That gap is the distance between the Stuart forced loans a few slides from now and the Crown that borrows at 14 percent in 1693 and 3 percent by 1739.

Skinner's is a claim about the concept. Strayer (1970) dates the machinery — permanent courts, records, regular taxation — to 1100–1300, and Davies (2003) doubts the category fits medieval polities at all. The defensible version is that impersonal and perpetual authority is what was new.

What English titles put after “state of”

What the English “state of ____” was a state of

Complement of the construction in English USTC titles. Before 1550 the word takes conditions of persons and souls. After 1600 it takes territories and institutions. The percentage is the polity share among the two substantive categories.



Source: USTC, 52,266 English editions 1450–1650; 1018 occurrences of “state(s)/estate(s) of”. Top twelve classified complements shown per period.

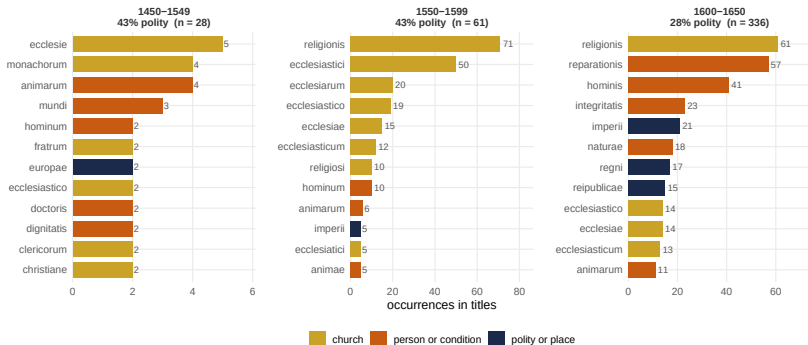
Category assignment is the authors’ coding and the full word lists are in the README. “Person or condition” includes the ruler’s own standing (status regis), which is the pre-modern sense. USTC records short titles, so this is title–page language rather than prose, and English coverage before 1550 is thin. One record dated 1483 that names Charles II is excluded as mis–dated; it was the only polity–coded complement before 1550.

Complements of “state(s)/estate(s) of” in English USTC titles, the top twelve of each period. Before 1550 the slot holds wedlock, matrimony, grace and childhood. After 1600 it holds kingdome, parliament, Scotland and Christendome. My own calculation from 52,266 English editions and 1,018 occurrences of the construction, 1450 to 1650.

Latin *status* keeps its genitive

Latin *status* keeps its genitive

Complement of *status*/*statu*/*statum*/*statui* in Latin USTC titles. The word stays attached to conditions of souls, persons and the church across all three periods, while the English cognate moves to territories.



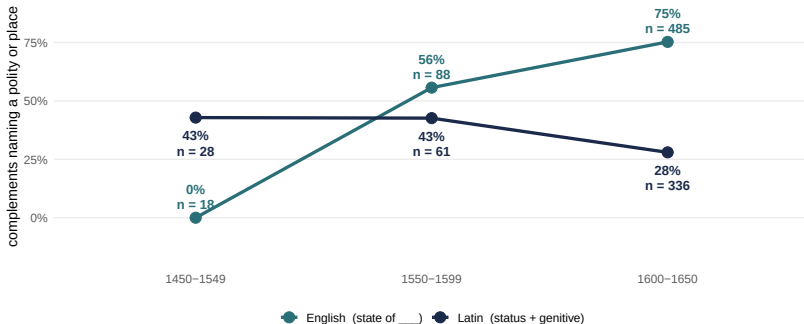
Source: USTC, 314,172 Latin editions 1450–1650; 1699 occurrences of *status* and its case forms. Top twelve classified complements shown per period. Category assignment is the authors' coding and the full word lists are in the README. *status* *reparationis*, *integritatis* and *corruptionis* are the technical theology of the states of man, and *status* *regni* or *reipublicae* remains the condition of a realm rather than an entity in itself.

The same exercise on *status* and its case forms in Latin titles. The slot keeps *religionis*, *ecclesie*, *animarum* and *hominum* across all three periods, and what grows after 1600 is the technical theology of the states of man. From 314,172 Latin editions and 1,699 occurrences. Latin is the control here, because it is the language the concept had to escape from.

The vernacular word moves and the Latin word stays put

The vernacular word moved and the Latin word did not

Share of the complement slot filled by a polity or place rather than by a person or condition. English crosses over in the second half of the sixteenth century, which is the window Skinner (2009) dates the modern concept to. Latin status stays a condition word throughout.



Source: USTC titles 1450–1650. Denominator is complements coded either polity or place or person or condition; church-coded and unclassified complements are excluded (English 18/29/38 percent unclassified by period, Latin 75/42/54 percent).

Category assignment is the authors' coding, with the full word lists in the README. Read the trends rather than the levels: the two baselines differ because pre-1550 English print is overwhelmingly devotional while Latin print of the same years is not.

English before 1550 rests on 18 classified occurrences and is suggestive rather than decisive.

Share of the complement naming a polity or place rather than a person or condition. English crosses over between 1550 and 1599, which is the window Skinner dates the concept to and the window that holds Bodin. Read the trends rather than the levels. The two baselines rest on different literatures, pre-1550 English print being overwhelmingly devotional, and the early English cell rests on 18 classified occurrences.

State capacity

State capacity, in two parts

Legal capacity. The state can enforce its rules across the entirety of the territory it claims to rule.

Fiscal capacity. The state can raise enough revenue from the economy to implement its policies.

Capacity is regularly confused with two other things.

- **Not size or scope.** The eighteenth-century British state had high capacity while playing a very limited role in the economy. A bloated public sector can be poor at both enforcing and collecting.
- **Not centralization.** Feudal lords devolved fiscal and legal choices downward and had low capacity. But concentrating everything in the center creates its own inefficiencies, and federalized states have done well.

This is a different axis from Week 3. Expropriation risk measured constraints on the state. Capacity measures what the state can actually do.

Four combinations of capacity and constraints

	Weakly constrained	Strongly constrained
High capacity	predation at scale (Stuart forced loans, sold monopolies)	the growth quadrant (Britain after 1688)
Low capacity	disorder, warlords (fragmented authority)	rules nobody enforces (paper constitutions)

- The Glorious Revolution settlement raised **both** at once. Constraints went up through parliamentary supremacy, independent judges and an exclusive parliamentary power to tax. Capacity went up through the Bank of England, funded debt and a far more effective tax administration.
- Most of history sits in one of the other three boxes. Getting both is the rare trick, and it takes more than doing either one well.

The predation quadrant

TABLE 2
FORCED LOANS BY THE EARLY STUARTS, 1603–1625

Year	Amount	Rate (percent per year)	Repayment
1604/5	£111,891	10%	£20,362 unpaid as of Dec. 1609
1611/2	116,381	10	£112,000 unpaid as of Jul. 1616
1617 ^a	96,466	10 ^b	Unpaid until 1628
1625 ^c	60,000	8	Unpaid until 1628

^a Extension in 1624 secured by Crown lands.

^b Unilaterally lowered by Charles I in 1624 to 8%.

^c Secured by Crown lands.

Source: Robert Ashton, *The Crown and the Money Market, 1603–1640* (Oxford, 1960), chaps. 2 and 5.

Forced loans levied by the early Stuarts. Note the last column. The 1604 loan was still partly unpaid five years later, the 1617 and 1625 loans were unpaid until 1628, and Charles I simply lowered the rate he owed from 10 to 8 percent. Source as shown, Ashton, *The Crown and the Money Market*.

What capacity buys

Take one example the survey works through. Trade on the **Rhine**.



- About **62 customs posts** on the river by the late middle ages. In the sixteenth century a merchant paid tolls at **31 points** between Basel and Cologne, roughly **one every nine miles**. Each tick above is one.
- Rates were not published. Most rulers kept their toll levels secret, and what was actually paid came out of negotiation between the boat captain and the toll collector on the spot.
- So every collector could price discriminate, and a merchant leaving Rotterdam could not know what he would be charged or where.

Nobody had to expropriate anybody for this to strangle trade. Fragmented authority did it, one toll at a time. A state strong enough to suppress local rent extraction is doing something a market cannot easily do for itself.

One collector on the whole river

Write demand for shipments as $Q = a - T$, where T is the sum of every toll paid between Basel and Cologne and a is the toll at which the river empties. Collecting costs nothing.

The merchant cares about the sum alone. Splitting a given T between two stations leaves his shipping decision exactly where it was, which makes the stations perfect complements. He has to pass every one of them.

Start with the easy case. One ruler owns the river end to end.

1. Revenue is the toll times the traffic it carries, $R = TQ = T(a - T)$.
2. Expanding gives $R = aT - T^2$.
3. Differentiating gives $\frac{dR}{dT} = a - 2T$.
4. Setting that to zero gives $T = a/2$. Traffic is $Q = a - a/2 = a/2$, and revenue is $a^2/4$.

That condition also reads $\frac{dR}{dT} = Q - T = 0$. Raising the toll earns him one more unit on the Q shipments that still come, and costs him T on the shipment he drives away. He stops where the two balance.

Two collectors on the same river

Now split the river in half. Collector 1 sets t_1 , collector 2 sets t_2 , and the merchant pays $T = t_1 + t_2$. Each takes the other's toll as given.

1. Collector 1 keeps only his own toll, so his revenue is $R_1 = t_1 Q = t_1 (a - t_1 - t_2)$.
2. Expanding gives $R_1 = at_1 - t_1^2 - t_1 t_2$.
3. Differentiating in t_1 gives $\frac{\partial R_1}{\partial t_1} = a - 2t_1 - t_2$.
4. Setting that to zero gives his **best reply**, $t_1 = \frac{a - t_2}{2}$. The more the other man charges, the less collector 1 charges.
5. Collector 2 solves the same problem, so $t_2 = \frac{a - t_1}{2}$.
6. Substituting the second into the first, $t_1 = \frac{a}{2} - \frac{a - t_1}{4} = \frac{a}{4} + \frac{t_1}{4}$, hence $\frac{3}{4}t_1 = \frac{a}{4}$ and $t_1 = t_2 = a/3$.

Each charges $a/3$, so the merchant pays $T = 2a/3$ and traffic falls to $Q = a/3$. Each collector takes $a^2/9$, and the two together take $2a^2/9$.

The term that drops out

Write both first-order conditions in terms of traffic.

- **One collector.** $Q - T = 0$, so he stops at $Q = T$. Every shipment he drives away costs him the whole toll.
- **Collector i of two.** $Q - t_i = 0$, so he stops at $Q = t_i$. Every shipment he drives away costs him his own toll alone.

The gap between the two conditions is t_j . The shipment collector i turns away was paying collector j as well, and that loss is absent from i 's problem. Each man therefore raises his toll past the point that would serve the two of them together.

	Two collectors	One collector
Total toll	$2a/3$	$a/2$
Traffic	$a/3$	$a/2$
Toll revenue	$2a^2/9$	$a^2/4$
Merchant surplus	$a^2/18$	$a^2/8$

The single collector charges less, moves more traffic and collects more revenue. Merchants and collectors both do better, so fragmentation destroys surplus rather than moving it from one pocket to another.

Thirty-one collectors

With n collectors the same algebra gives $t_i = \frac{a}{n+1}$, so $T = \frac{na}{n+1}$ and

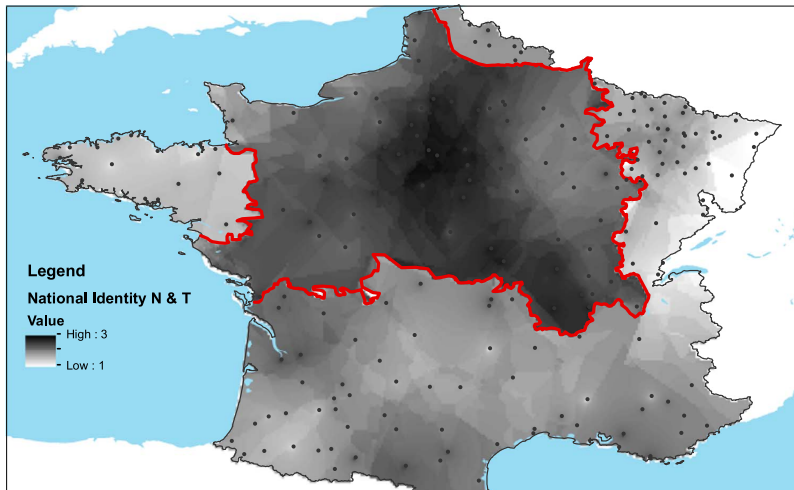
$Q = \frac{a}{n+1}$. Joint revenue is $\frac{na^2}{(n+1)^2}$.

Collectors	Total toll	Traffic	Joint revenue
1	$a/2$	$a/2$	$0.250 a^2$
2	$2a/3$	$a/3$	$0.222 a^2$
5	$5a/6$	$a/6$	$0.139 a^2$
31	$31a/32$	$a/32$	$0.030 a^2$

Thirty-one is the number of toll points a merchant actually met between Basel and Cologne. Traffic falls to a thirty-second of the level that would empty the river, and the collectors together take about an eighth of what one of them would take alone.

Cournot (1838) on complementary monopoly. Shleifer and Vishny (1993) give the modern treatment, aimed at independent takers of bribes rather than tolls.

Fiscal and legal capacity may also interact...



National identity in the 1789 General Cahiers, darker meaning stronger identification with the king or with France. The red line is the Cinq Grosses Fermes, the internal customs union. Taxes collected per capita ran about 40 percent higher inside it. Johnson and Koyama (2017), Figure 2, from Johnson (2015).

And what the constraints bought

TABLE 4
GOVERNMENT LONG-TERM BORROWING: INTEREST RATES, 1693–1739
(selected loans)

Date ^a	Amount	Interest	How Funded
Jan 1693	£723,394	14.0%	Additional excise
Mar 1694	1,000,000	14.0	Duties on imports
Mar 1694	1,200,000	8.0	Additional customs and duties
Apr 1697	1,400,000	6.3	Excise and duties
Jul 1698	2,000,000	8.0	Additional excise duties
Mar 1707	1,155,000	6.25	Surplus from funds of five loans from 1690s; duties
Jul 1721	500,000	5.0	Hereditary revenue of Crown
Mar 1728	1,750,000	4.0	Coal duties
May 1731	800,000	3.0	Duties
Jun 1739	300,000	3.0	Sinking fund

^a Date of royal assent to loan act.

- The Crown borrows at **14 percent** in 1693 and at **3 percent** by 1739, on far larger sums, from lenders who had watched the Stuarts default within living memory.
- Nothing about the technology of lending changed. What changed was who controlled the purse and what happened to a king who reneged.

Read the claim carefully

Their phrase is **strong but limited states**. Both at once, which is much harder and much rarer than either alone.

- They are careful, and you should be too. The link between capacity and growth is a **contingent** one.
- Their own account puts the origins of growth in the expansion of market exchange and in cultural change. What high-capacity states did was supply conditions that let growth happen, or at least kept warfare and rent-seeking from destroying it.
- Compare Korea with the Philippines. Similar GDP in 1960, and utterly different histories of statehood behind them.

Notice this is the Federalist 10 problem again, from the other side. Madison wanted enforcement power that no faction could capture, so he built it and then split it.

Reallocation and the Coase theorem

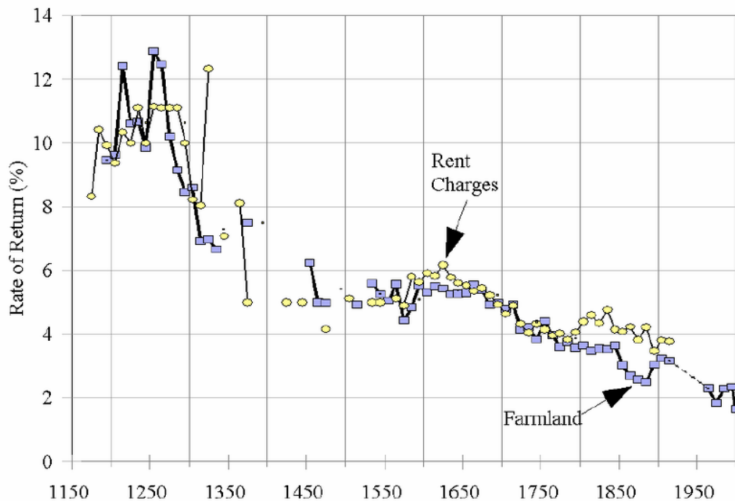
A complication about 1688

The standard account, from North and Weingast, is that the Glorious Revolution made property rights **secure**, and that security is what unlocked growth.

- Pincus and Robinson argue that North and Weingast were **right for the wrong reasons**.
- What happened in 1688 was a **reallocation** of property rights, from one set of interests to another. Whigs over Tories, trade over agriculture, urban over rural, war finance and public debt over low taxes.
- Property rights were not obviously insecure in England before 1688. Some rights changed hands, and the ones that ended up in charge were the ones that wanted infrastructure, a navy and a funded debt.

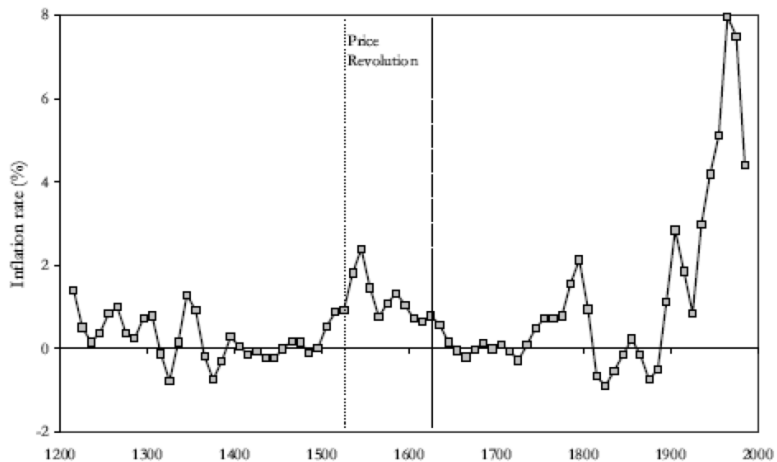
If growth follows from which interests hold the rights rather than from how firmly they are held, then we need a framework in which the initial allocation matters. That framework is the Coase theorem, and it is almost always taught backwards.

Look for the break



Rate of return on rent charges and on farmland, 1150 to 1950. Clark's series. Returns fall for eight centuries and 1688 is not visible in them. If the Glorious Revolution had suddenly made property safe, this is where it would show.

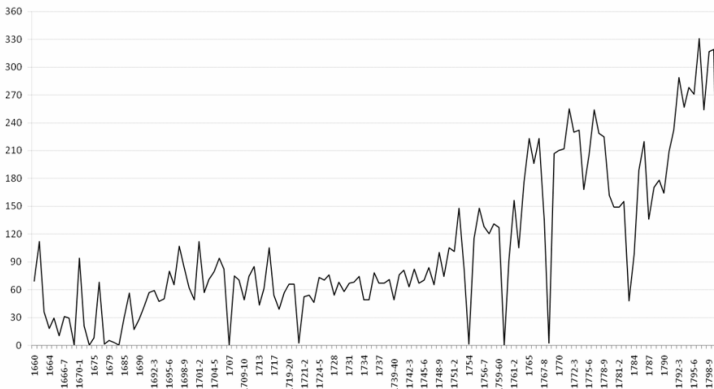
Nor is it in prices



Inflation rate, 1200 to 2000. The Price Revolution of the sixteenth century is plainly there. 1688 is not.

What did change — Parliament started legislating

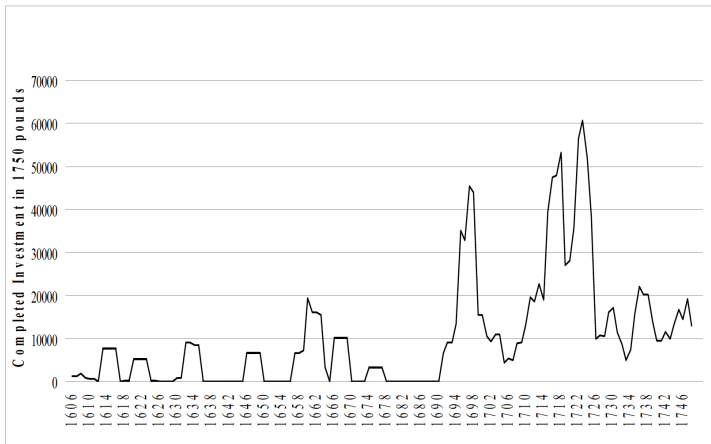
Figure 2: Amount of Parliamentary Legislation per year, 1660-1798



Parliament began legislating at a rate with no precedent, and much of that legislation reorganized who held what. Pincus and Robinson, Figure 2, as reproduced in the 2025 lecture deck.

And the money went into infrastructure

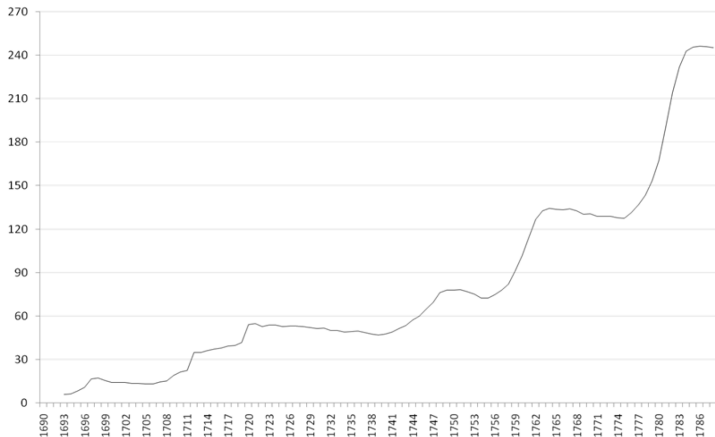
Figure 3: Four-Year Moving Average of Completed Investment in Road and River Improvements, 1607-1749



Completed investment in road and river improvements, four-year moving average. It takes off in the 1690s and never returns to the old level. Pincus and Robinson, Figure 3.

And the borrowing that paid for it

Figure 5: Total Outstanding Government Debt (millions of pounds), 1690 to 1788.



Total outstanding government debt, 1690 to 1788, in millions of pounds. From almost nothing to about 250 million. This is the same fact as the falling interest rate two sections ago, seen from the other side. Pincus and Robinson, Figure 5.

Reallocation to a coalition that would build

Put the last four slides together. Property rights were not obviously less secure before 1688, and nothing breaks in 1688. What 1688 did was hand power to the **Whigs** — war, trade, towns, infrastructure, high taxes and public debt — and away from the Tories, who wanted peace, agriculture, low taxes and no debt.

- Transaction costs in seventeenth-century England were enormous. Feudal and customary tenures could not simply be bargained away, and entails could not be undone by private agreement.
- So Parliament did it by statute, thousands of times over, dismantling preexisting rights in favor of rights suited to commercial agriculture and to industry.
- Johnson and Koyama make the same point about the same episode. The state after 1688 **reordered property rights** more than it increased protection for the existing ones.

The Coase theorem, as it is usually taught

The textbook statement

In a world of zero transaction costs, property rights will be traded until an efficient allocation is reached, **regardless of the initial allocation of property rights**.

- This is the version in the managerial economics textbooks, and it is attributed to Coase by way of **Stigler**, who named it.
- It is routinely deployed to argue that the state should stay out of markets. If bargaining fixes everything, the law is a starting point that washes out.



George Stigler, who named the theorem, and Ronald Coase, who wrote the paper it is named after. They did not mean the same thing by it.



More trouble brewing

A skydiving school and a crocodile farm

Suppose operating is worth \$100 to the skydiving school and \$50 to the crocodile farm, and the farm imposes an externality of \$75 on the school.

Outcome	Total surplus
Both operate	$100 + 50 - 75 = 75$
School only	100
Farm only	50

- **Liability lies with the farm.** The school forces it to shut. Surplus is 100.
- **Liability lies with the school.** The school pays the farm \$50 to shut. Surplus is still 100, now split \$50 and \$50.

The farm shuts either way. The initial allocation changed only who got the money. That is the result everyone remembers, and it holds only because the bargaining and the injunction were both assumed to be costless.

What Coase actually said

The bargain in case two requires transaction costs low enough to strike it. The injunction in case one requires courts cheap enough to reach. Neither is free.

"I tend to regard the Coase Theorem as a stepping stone on the way to an analysis of an economy with positive transaction costs... If we move from a regime of zero transaction costs to one of positive transaction costs, what becomes immediately clear is the crucial importance of the legal system in this new world... It makes little sense for economists to discuss the process of exchange without specifying the institutional setting within which the trading takes place since this affects the incentives to produce and the costs of transacting."

The zero-transaction-cost world was the **setup**. Coase raised it in order to focus attention on the institutions that affect transaction costs. In the world we actually live in, the initial allocation of rights matters enormously, and so does the machinery that reassigns them.

Coase, Nobel Prize lecture, 1991.

Why the misreading has mattered

- **Regulatory policy.** Public goods, environmental rules, financial regulation, net neutrality — all argued at some point on the grounds that private bargaining would sort it out.
- **Development policy.** The Washington Consensus, post-Soviet privatization, and the “political Coase theorem” that efficient institutions will be adopted because they are efficient.
- **Market design.** Spectrum auctions, oil field auctions, emissions trading. Here the Stiglerian reading did real good, because these are cases where transaction costs were deliberately engineered down.

Hold that last bullet next to the other two. The lesson is that whether the initial allocation washes out is an **empirical question about transaction costs**, to be asked case by case.

The mystery of property rights

An anomaly

Everyone agrees that secure property rights drive development. Lamoreaux does not dispute it. She points at a set of facts that do not fit.

- American history is a long parade of **involuntary reallocations** of property. Charters overridden, land taken, water diverted, entitlements rewritten.
- And the United States grew anyway. Spectacularly, and often precisely in the sectors and regions where the taking happened.
- She calls these facts **anomalies** in Kuhn's sense. They are a sign the theory is missing something.

Week 3's expropriation-risk index would score much of this as insecurity. Hold on to that thought.

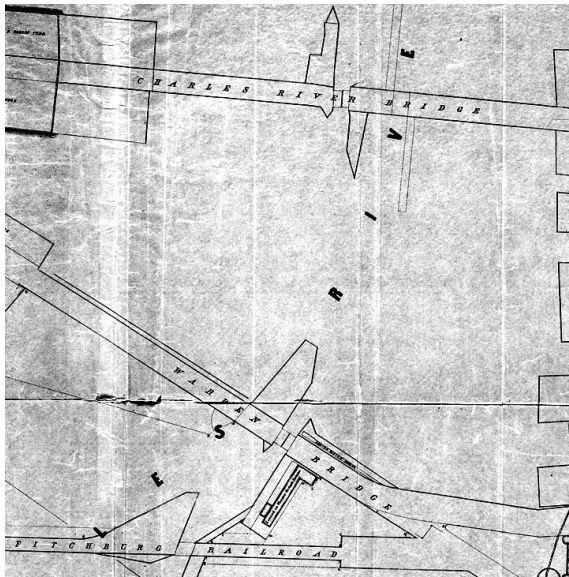
Lamoreaux (2011), "The Mystery of Property Rights: A U.S. Perspective."

The Charles River Bridge

- **1785.** Massachusetts charters private investors to bridge the Charles at Charlestown. Nobody knows whether it can be built at all, so the legislature grants forty years of tolls to compensate the risk. Later extended to seventy.
- The bridge is built. It is very profitable, and in being profitable it demonstrates that such bridges can be built.
- **1828.** The legislature charters a second bridge alongside it, which collects tolls for six years and then becomes free.
- The original proprietors sue. They lose in state court, and they lose in the Supreme Court in **1837**.

Justice Story, dissenting, asks whether any man of ordinary prudence would “hazard his capital” if the legislature could later “destroy his profits, and annihilate his tolls.” If government wants investment, “there must be some pledge, that the property will be safe.”

The two bridges



The Charles River Bridge of 1785 and, a few hundred yards downstream, the Warren Bridge that was chartered in 1828 to compete with it and then made free. Plan of 1854, Norman B. Leventhal Map Center, Boston Public Library, CC BY 2.0.

And yet

Story's logic is the standard logic, and it is not wrong. But look at what followed.

- Investment did not collapse. Massachusetts industrialized.
- **The mill acts** went further, handing powers of eminent domain to private manufacturers. Upstream landowners were flooded out so that mills downstream could raise their dams, with compensation set by schedule rather than by consent.
- Riparian rights were rewritten to favor industrial use over the older agricultural claims that shared the same water.

This is involuntary reallocation as **growth policy**, and it worked. Read through Coase, that is exactly what you would expect. With positive transaction costs, the state that reassigns the right is doing productive work that bargaining would not have done.

Two takings, two centuries later

	Midkiff, 1984	Kelo, 2005
What was taken	Hawaiian land oligopoly	a private house
Given to	existing tenants	a redevelopment project
Court	unanimous	split 5-4
Public reaction	almost none	national outrage

- The same doctrine, applied twice. In 1984 not one justice dissented from the view that breaking up concentrated landholding was a legitimate public purpose, and the press barely covered it.
- In 2005 a far more routine urban redevelopment produced a 5-4 decision and a national campaign.

So the constraint on takings is political as much as doctrinal. What changed between these two cases was who felt threatened.

How Lamoreaux resolves it

Her answer is that property was **widely held**.

Because most voters owned property, they could safely grant government the authority to rearrange property rights for the common good — that is, they could let property rights be **flexible**. And they could protect themselves against abuse, because if powerful interests captured the machinery, voters could take back the authority they had given up.

- And they did take it back, repeatedly. New York wrote compensation by jury into its constitution in **1821**, and in **1846** required a jury of freeholders to find a private road necessary before it could be built. Ohio in **1851** required compensation assessed by jury with no deduction for benefits to the owner's remaining land.
- She names the dark side herself. Because property was so widely held, **redistribution toward the bottom was kept in check**. Any serious reallocation to benefit the propertyless would hurt too many voters.

What that does to Week 3

- Security and flexibility are **both** required, and they pull against each other. A polity that can never take is a polity that can never build a road or a mill or a railway.
- What resolves the tension is a **distribution**. The protection came from the fact that the people who could be taken from were also the people who could vote.
- Notice this is Federalist 10 running in reverse. Madison feared the propertied minority being outvoted. Lamoreaux describes a propertied **majority** that is safe for exactly that reason, and a propertyless minority that is not.

An oral exam question, so think about it now. Acemoglu, Johnson and Robinson used protection against expropriation to measure institutional quality. How much of what Lamoreaux describes would that index score as **bad institutions**?

The rules people carry

Turner's thesis, 1893

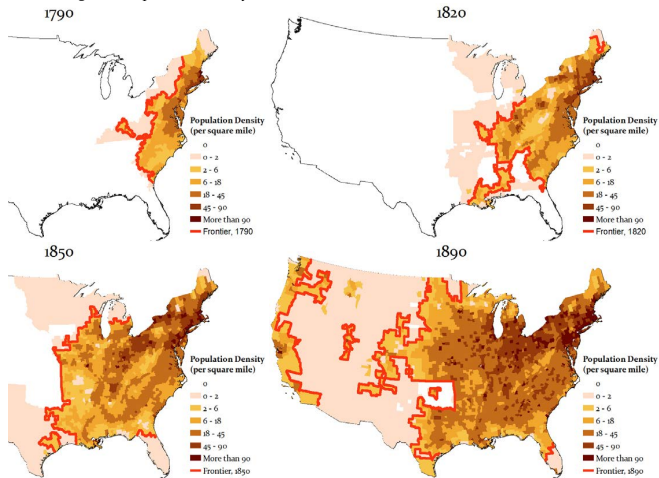
- Frederick Jackson Turner argued that the American frontier was the crucible of American character. Self-reliance, suspicion of authority, individualism.
- It is a claim about **culture** — the informal half of the institutions definition we started Week 3 with.
- It is also, for a century, exactly the kind of claim economists could not do anything with. Bazzi, Fiszbein and Gebresilasse take it to census microdata and a GIS.

Ask the question before you see the answer. If the frontier really did shape American culture, what would have to show up in data, and where would you look for it?

Bazzi, Fiszbein and Gebresilasse, "Frontier Culture." Numbers here are from the 2017 working paper. The syllabus cites the 2020 Econometrica version.

The frontier line moves west

Figure 1: Population Density and the Frontier for Selected Years



Bazzi, Fiszbein and Gebresilasse (2017 working paper), Figure 1. County population density with the frontier line in red. The frontier is the contour at two people per square mile.

Building the treatment variable

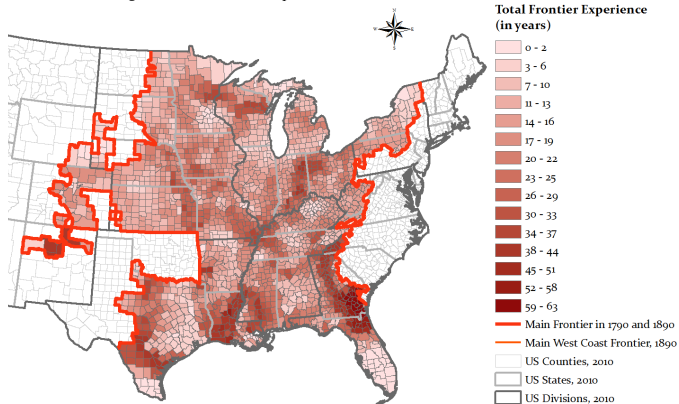
Total frontier experience

For each county, the number of years between 1790 and 1890 that it sat within 100 kilometers of the frontier line while its own population density was below six people per square mile.

- It ranges from **0 to 63 years**, with a mean of about 18 and a standard deviation of about 11. Counties a short distance apart can differ a great deal, because the frontier swept past some places and lingered at others.
- Frontier counties look distinctive while they are frontier. A sex ratio of about **1.09**, more prime-age adults, more illiterate residents, more foreign-born.
- Individualism is proxied by the share of children under ten given **infrequent names**. Validated against survey measures of individualism, and still only a proxy.

Total frontier experience by county

Figure 3: Total Frontier Experience, 1790 to 1890



Bazzi, Fiszbein and Gebresilas (2017 working paper), Figure 3. Years of frontier experience, 1790 to 1890. This is the treatment variable itself.

Their specification

$$y_c = \alpha + \beta \text{TFE}_c + x'_c \gamma + \theta_{FE(c)} + \varepsilon_c$$

- A cross-section of counties. y_c is an outcome capturing individualism or preferences over government, and TFE is measured in **decades**.
- x_c holds latitude, longitude, county area, rainfall, temperature, elevation, predetermined agricultural yield, and distance to rivers, lakes and the coast. θ is state or Census-division fixed effects.
- Standard errors are clustered on **60-square-mile grid squares** rather than on counties, following Bester, Conley and Hansen.

That clustering method is Bester, **Conley** and Hansen's — the same Timothy Conley whose paper closes Week 3. And Conley and Kelly use that very estimator as their own workhorse.

So the two papers are using the **same tool**, and it still is not enough. What Conley and Kelly add is the spatial basis that removes the trend first.

This is a cross-section with controls, and it is exactly the design Conley and Kelly built their critique around.

Frontier experience and six outcomes

Table 3: Total Frontier Experience and Opposition to Government Intervention and Redistribution

Dependent Variable:	Prefers Cut Public Spending on Poor binary	Prefers Cut Public Spending on Welfare binary	Believes Gov't Should Redistribute normalized	Prefers Reduce Debt by Spending Cuts binary	Index of Preferences for Spending Cuts normalized	County Property Tax Rate, 2010 [0, 100]
Scale:	ANES	CCES	GSS	CCES	GSS	ACS
Data Source:	(1)	(2)	(3)	(4)	(5)	(6)
total frontier experience	0.010*** (0.004)	0.007** (0.003)	-0.022* (0.012)	0.014*** (0.002)	0.028** (0.011)	-0.034*** (0.007)
Oster δ for $\beta = 0$	5.59	6.86	5.79	2.40	2.28	1.67
Mean of Dependent Variable	0.09	0.40	0.00	0.43	-0.00	1.02
Number of Individuals	2,322	53,472	9,085	169,630	5,739	–
Number of Counties	95	1,863	255	2,001	253	2,029
R ²	0.04	0.04	0.06	0.04	0.07	0.82
Survey Wave Fixed Effects	Yes	Yes	Yes	Yes	Yes	–
Individual Demographic Controls	Yes	Yes	Yes	Yes	Yes	–
State Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Geographic/Agroclimatic Controls	Yes	Yes	Yes	Yes	Yes	Yes

Bazzi, Fiszbein and Gebresilasase, Table 3. Six outcomes, from stated preferences over public spending in three different surveys to the actual county property tax rate in 2010. Every coefficient runs the direction the thesis predicts.

Two ways of defending a cross-section

Bazzi and co-authors answer the objection twice.

- **Oster bounds.** The row marked δ asks how large selection on **unobservables** would have to be, relative to the observables already controlled for, to drive the coefficient to zero. Values of **5.6** and **6.9** in the first two columns mean the unmeasured confounders would need to be roughly six times as powerful as everything measured. That is a hard story to tell.
- **An instrument.** A shift-share of the kind Bartik made standard. National immigration inflows pushed the frontier west faster in some decades than others, and that timing is plausibly unrelated to any one county's later culture.

The Oster bound and the instrument attack the same problem from opposite ends. One asks how bad the omitted variable would have to be. The other tries to find variation the omitted variable cannot touch.

Building the instrument

A shift-share multiplies a **national shock** nobody local controls by a **local exposure** fixed before that shock arrives.

- **The exposure.** For each county, the first year the frontier line stands 110 kilometers to its east. The county is still unsettled then, so its own conditions cannot be moving the wave.
- **The shock.** Average annual immigration into the United States over the next 30 years. Nearly 85 percent of counties leave the frontier inside that window.
- **The logic.** Heavy inflows drive the wave through faster, and a faster wave leaves less frontier experience behind it.

What has to be true

The frontier line's position in a given year is unaffected by conditions more than 100 kilometers farther west, and the volume of immigration into the country is unaffected by the culture of any one frontier county.

The second is the fragile one. If individualists in Europe sailed because they had heard the frontier land was good, that inflow is a **pull** factor and the condition fails. So they rebuild the instrument from **push** alone, using European climate shocks to predict outflows, and the estimates hold.

The instrumental variables estimates

Dependent Variable:	Infrequent Name Share normalized		County Property Tax Rate	Republican Vote Share Avg. 2000–16
	Boys 1940 (1)	Girls 1940 (2)	2010 (3)	(4)
Panel A: Baseline Sample and Specification				
total frontier experience	0.121** (0.051)	0.113** (0.057)	-0.035*** (0.014)	1.492** (0.715)
Number of Counties	2,036	2,036	2,029	2,036
Mean of Dependent Variable	0.00	0.00	1.02	60.04
KP Wald Statistic	216.3	216.3	216.1	216.3
Panel B: Adding Controls for Population Density Birthplace Diversity, and Initial Railroad Access				
total frontier experience	0.154*** (0.043)	0.191*** (0.051)	-0.045*** (0.012)	1.797*** (0.655)
Number of Counties	1,839	1,839	1,837	1,839
Mean of Dependent Variable	0.00	0.00	1.03	59.57
KP Wald Statistic	245.0	245.0	244.8	245.0
State Fixed Effects	Yes	Yes	Yes	Yes
Geographic/Agroclimatic Controls	Yes	Yes	Yes	Yes

Table 7. Kleibergen–Paap above 200 says the instrument is strong. Panel B exceeds its Table 6 OLS counterpart, though the authors call Panel A indistinguishable from OLS.

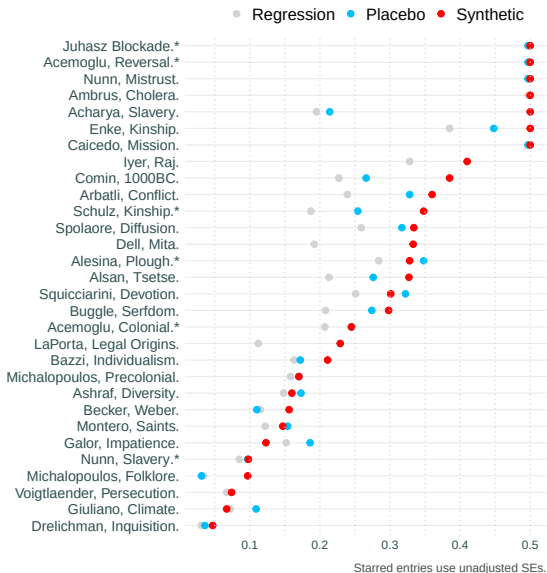
What persists

Decades after the frontier closed, counties with more frontier experience show

- more individualism, on the naming measure, generations later;
- more opposition to redistribution and regulation, across several different surveys;
- **lower local property tax rates**, which is a policy outcome rather than a stated preference;
- and about **3.5 percent** more votes for Republican presidential candidates per decade of frontier experience, in elections since 2000.
- For identification they instrument with the **speed of westward expansion** driven by national immigration inflows, and bound the selection on unobservables using Oster's method.

Two mechanisms, and they find both. **Selection**, because individualists chose the frontier. And **treatment**, because frontier conditions rewarded self-reliance, which they trace in linked census records following individuals.

You have seen this figure before



Conley and Kelly (2025), Figure 7, from Week 3. “Bazzi, Individualism.” is on the list, and it does not clear 5 percent under their spatial basis procedure either.

Reading a paper with both eyes open

- Frontier experience is a **map**. Individualism today is a map. Everything Conley and Kelly said last week applies here, and this paper is in their sample.
- So do not leave this room believing the 3.5 percent is a precisely estimated causal quantity. Under honest spatial inference it is not distinguishable from a trend.
- **But be equally careful in the other direction.** A great deal of this paper is not a persistence regression at all. The historical demography is measurement. The linked-census evidence on who moved and what happened to them is not a cross-sectional map correlation, and it does not fall with the persistence coefficient.

Hold the substantive claim and the inferential critique in your head **at the same time**, and be precise about which parts of a paper each one touches.

Where this leaves us

Are culture and institutions separable?

Exhibit	What kind of rule	Who enforces it
Federalist 10	designed, formal	the structure itself
Johnson and Koyama	enforced, formal	a state, if it can
Lamoreaux	reallocated, formal	propertied voters
Bazzi et al.	internalized, informal	the people who hold it

- They are probably not separable. Each row generates and stabilizes the others. Madison's design assumed a certain kind of citizen. Lamoreaux's mechanism runs on who owns what. Frontier culture is partly a response to weak state capacity at the edge of settlement.
- That is an **empirical** constraint. You cannot instrument for institutions while holding culture fixed if the two co-evolve, and much of the literature quietly assumes you can.

Four readings and four kinds of evidence

	What it is	How it persuades
Federalist 10	a primary source	argument from first principles
Johnson and Koyama	a survey	synthesis of others' evidence
Lamoreaux	legal history	cases, statutes, constitutions
Bazzi et al.	an empirical paper	coefficients and standard errors

- Only the last one has a regression in it, and that is not an accident of who was assigned. **Three of tonight's four questions cannot be answered with a regression.**
- You cannot run a specification on what Madison believed, on what a thousand years of state formation add up to, or on whether a nineteenth-century court was right. Those need reading, synthesis and judgment.
- The cost is that Lamoreaux cannot put a standard error on her claim, so **reasonable people can still disagree about it.**

An economic historian has to be fluent in all four registers. The orals will ask you to move between them.

How to read an institutions paper

The companion to last week's list. These get asked in the oral exams too.

1. **Which rule?** Formal or informal, written or carried in the head?
2. **Who enforces it,** and what does enforcement cost?
3. **Secure for whom, against whom?** Name both parties.
4. **Is this insecurity or flexibility?** Would Lamoreaux read the paper's "bad institution" as a functioning reallocation?
5. **Are transaction costs doing work?** If the initial allocation matters, say why bargaining did not undo it.
6. **If culture is the channel,** what pins it down, and did the standard errors survive Week 3?

Question four is the one this week added. Photograph this.

Next week

- **Week 5, 28 September. Industrialization.**
- Allen on why the industrial revolution happened in Britain, and whether high wages and cheap coal are enough of an answer.
- Clark on why the whole world is not developed, which is the same question pointed at the present.
- Hanlon on necessity and invention, and whether the direction of technical change responds to prices.

We now have designed rules, states that can enforce them, courts that reallocate them, and cultures that carry them. Next week asks why the growth engine itself started where and when it did.